

Question 1

[Maximum mark: 5]

Consider the following statement.

“The product of any two consecutive even integers is divisible by 8.”

(a) Show that this statement is true for the integers 12 and 14 [2]

(b) Prove that the statement is true for any two consecutive even integers. [3]

Answer:

(a) Product = $12 \times 14 = 168$

Which is divisible by 8.

(b) Let two consecutive even integers are: $2k, (2k + 2)$

$$\therefore \text{Product} = (2k)(2k + 2) = 4k^2 + 4k = 4k(k + 1) \quad \text{---(i)}$$

Now k and $(k+1)$ are two consecutive integers, one of which is even, therefore, their product is even,

Let

$$2m = k(k + 1), m \in \mathbb{Z}$$

$$\text{Product} = 4(2m) = 8m$$

which is divisible by 8.

[Maximum mark: 6]

Consider two events, A and B , such that

$$P(A) = x, \quad P(B) = 5x - 1, \quad P(A \cup B) = 0.65.$$

(a) Find an expression for $P(A \cap B)$. [2](b) Given that events A and B are independent, find the value of x . [4]**Answer:**

SL 4.6	Combined events	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
-----------	-----------------	---

$$(a) \quad P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= x + (5x - 1) - 0.65$$

$$P(A \cap B) = 6x - 1.65 \quad \text{———— (i)}$$

(b)

	Independent events	$P(A \cap B) = P(A)P(B)$
--	--------------------	--------------------------

$$P(A \cap B) = P(A) \times P(B)$$

from (i)

$$P(A) \times P(B) = 6x - 1.65$$

$$\therefore (x)(5x - 1) = 6x - 1.65$$

$$5x^2 - x = 6x - 1.65$$

$$\therefore 5x^2 - 7x + 1.65 = 0$$

$$x = \frac{7 \pm \sqrt{7^2 - 4(5)(1.65)}}{2(5)} = \frac{7 \pm 4}{10}$$

$$\therefore x = \frac{11}{10}, x = \frac{3}{10}$$

$$\text{but } x \text{ is probability} \Rightarrow x \leq 1$$

$$(0 \leq P(A) \leq 1)$$

$$\therefore x = \frac{3}{10}$$

Question 3

[Maximum mark: 6]



(a) Express $\log_2 \left(\frac{3x^4}{y} \right)$ in terms of $\log_2 x$ and $\log_2 y$. [3]

(b) Hence, given $y = x^2$, solve $\log_2 \left(\frac{3x^4}{y} \right) = 3$. [3]

Answer:

$$(a) \quad \log_2 \left(\frac{3x^4}{y} \right) = \log_2 3 + 4 \log_2 x - \log_2 y$$

$$(b) \quad y = x^2$$

$$\therefore \log_2 \left(\frac{3x^4}{y} \right) = 3 \Rightarrow \log_2 \left(\frac{3y^2}{y} \right) = 3$$

$$\Rightarrow \log_2 (3y) = 3$$

$$\Rightarrow 3y = 2^3$$

$$\therefore y = \frac{8}{3}$$

Question 4

[Maximum mark: 6]



Consider the function $f(x) = x^2 - 6x + 10$

(a) (i) Express $f(x)$ in the form $f(x) = (x - h)^2 + k, k \in R$

(ii) Hence write down the coordinates of the vertex of the graph of f . [2]

(b) The line $y = 2x + c$ is a tangent to the graph of f . Find the value of c . [3]

(c) Hence find the coordinates of the point where the line touches the graph of f . [1]

Answer:

(a) (i) $f(x) = (x - 3)^2 - 9 + 10$

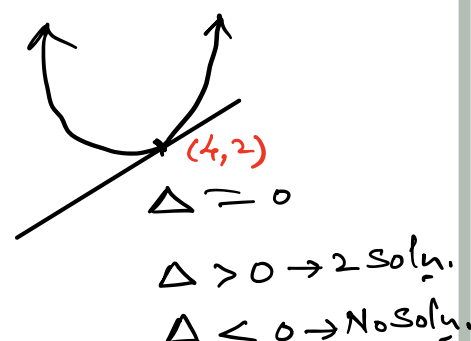
$$\therefore f(x) = (x - 3)^2 + 1$$

(ii) vertex is $(3, 1)$

(b) Solving line and curve, we get

$$x^2 - 6x + 10 = 2x + C$$

$$x^2 - 8x + (10 - c) = 0 \quad \text{--- (i)}$$



Since line is tangent to curve,

$$\Delta = 0$$

$$\therefore b^2 - 4ac = 0$$

$$(-8)^2 - 4(1)(10 - c) = 0$$

$$64 - 40 + 4c = 0$$

$$c = -\frac{24}{4} = -6$$

(C) From equation (i), we get

$$x^2 - 8x + 10 - (-6) = 0$$

$$x^2 - 8x + 16 = 0$$

$$\therefore (x - 4)^2 = 0$$

(always bracket square)

$$\therefore x = 4$$

$$\text{and } y = 2x + (-6)$$

$$\therefore y = 2(4) - 6 = 2$$

Therefore, line touches $f(x)$ at $(4, 2)$

Question 5

[Maximum mark: 7]



(a) Show that $4 - 3 \cos 2x = 6 \sin^2 x + 1$. [1]

(b) Hence or otherwise solve $4 - 3 \cos(4\theta + \frac{2\pi}{3}) - 9 \sin(2\theta + \frac{\pi}{3}) = -2$ for $0 \leq \theta < \pi$. [6]

Answer:

$$\begin{aligned} (a) \quad & 4 - 3 \cos 2x \\ &= 4 - 3(1 - 2 \sin^2 x) \\ &= 1 + 6 \sin^2 x \\ &= 6 \sin^2 x + 1 \end{aligned}$$

$$\begin{aligned} \cos 2x &= \cos^2 x - \sin^2 x \\ &= 2 \cos^2 x - 1 \\ &= 1 - 2 \sin^2 x \end{aligned}$$

(b) $4 - 3 \cos(4\theta + \frac{2\pi}{3}) - 9 \sin(2\theta + \frac{\pi}{3}) = -2$ for $0 \leq \theta < \pi$.

Let $x = 2\theta + \frac{\pi}{3}$

$$\therefore 4 - 3 \cos 2x - 9 \sin x = -2$$

$$(6 \sin^2 x + 1) - 9 \sin x = -2$$

$$\therefore 6 \sin^2 x - 9 \sin x + 3 = 0$$

$$2 \sin^2 x - 3 \sin x + 1 = 0$$

$$\therefore \sin x = \frac{3 \pm \sqrt{9 - 4(2)(1)}}{2(2)} = \frac{3 \pm 1}{4}$$

$$\therefore \sin x = 1 \quad \text{and} \quad \sin x = \frac{1}{2}$$

$$\Rightarrow x = \frac{\pi}{2} \quad \text{and} \quad x = \frac{\pi}{6}, \frac{5\pi}{6}$$

$$\rightarrow \therefore 2\theta + \frac{\pi}{3} = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{12}$$

$$\rightarrow 2\theta + \frac{\pi}{3} = \frac{\pi}{6} \Rightarrow \theta = -\frac{\pi}{12}$$

But $-\frac{\pi}{12}$ is out of range of $0 \leq \theta < \pi$

We add or subtract 2π in it

$$\therefore 2\theta + \frac{\pi}{3} = \frac{\pi}{6} + 2\pi = \frac{13\pi}{6}$$

$$\rightarrow \therefore \theta = \frac{11\pi}{12}$$

$$\rightarrow 2\theta + \frac{\pi}{3} = \frac{5\pi}{6} \Rightarrow \theta = \frac{\pi}{4}$$

$$\therefore \theta = \frac{\pi}{12}, \frac{\pi}{4}, \frac{11\pi}{12}$$

Question 6

[Maximum mark: 7]



The function f is defined by

$$f(x) = \frac{x^2 + 4}{x}, \quad x \in \mathbb{R}, x \neq 0.$$

- (a) Find the equations of the asymptotes of the graph of f . [2]
- (b) Sketch the graph of $y = f(x)$, showing the asymptotes and the coordinates of any local maximum or local minimum points. [3]
- (c) Hence find the set of values of k for which the equation $|f(x)| = k$ has exactly four solutions. [2]

Answer:

$$(a) \quad f(x) = \frac{x^2 + 4}{x} = x + \frac{4}{x}$$

$$\text{as } x \rightarrow \pm\infty, \frac{4}{x} \rightarrow 0$$

$\therefore y = x$ is oblique asymptote

Also denominator is zero at $x = 0$ is vertical asymptote

$$\therefore x = 0$$

$$(b) \quad f(x) = x + \frac{4}{x}$$

$$f'(x) = 1 - \frac{4}{x^2}$$

$$f'(x) = 0 \Rightarrow 1 - \frac{4}{x^2} = 0$$

$$\therefore x^2 = 4$$

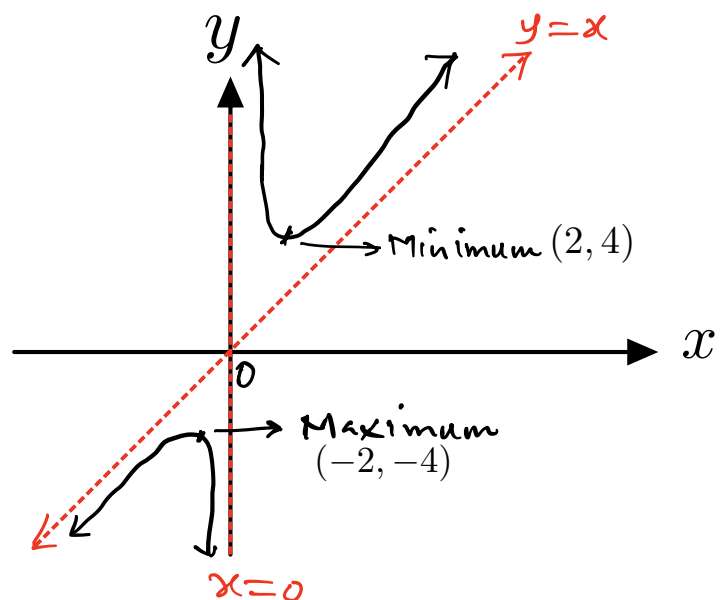
$$\Rightarrow x = \pm 2$$

$$f(2) = 2 + \frac{4}{2} = 4$$

$$f(-2) = -2 + \frac{4}{-2} = -4$$

$(2, 4)$ is a minimum point

$(-2, -4)$ is a maximum point



(c)

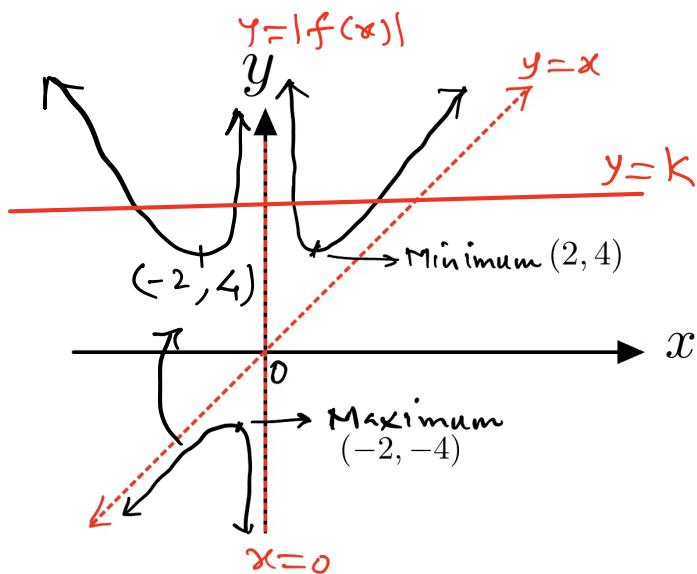
(c) Hence find the set of values of k for which the equation $|f(x)| = k$ has exactly four solutions.



$$\left| \frac{x^2 + 4}{x} \right| = k$$

Using property of absolute function, above graph will be above x-axis

This graph has 4 solutions for $k > 4$



Question 7

[Maximum mark: 7]



The curve S has equation $y^2 - x^2 = 5$.

(a) Show that the normal to S at (2, 3) is $y = -\frac{3}{2}x + 6$ [3]

The region bounded by the coordinate axes and this normal is rotated 2π radians about the y-axis.

(b) Find the resulting volume. [4]

Answer:

(a) Differentiating with respect to 'x', we get

$$2y \cdot \frac{dy}{dx} - 2x = 0$$

$$\frac{dy}{dx} = \frac{x}{y}$$

$$\text{at } (2, 3), \frac{dy}{dx} = \frac{2}{3} = m_1$$

is gradient of tangent,

Gradient of normal is

$$m_2 = -\frac{3}{2}$$

$\therefore$ equation of normal at $(x_1, y_1) = (2, 3)$ is

$$y - y_1 = m_2(x - x_1)$$

$$y - 3 = \frac{-3}{2}(x - 2)$$

$$\therefore y = -\frac{3}{2}x + 3 + 3$$

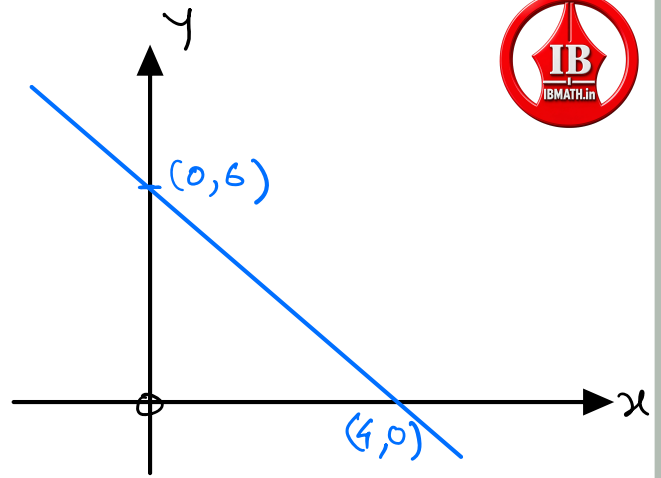
$$y = -\frac{3}{2}x + 6$$

(b) From equation of normal

$$\frac{3}{2}x = 6 - y$$

$$x = \frac{2}{3}(6 - y)$$

$$\therefore x^2 = \frac{4}{9}(6 - y)^2$$



$$Volume = \pi \int_0^6 x^2 dy$$

$$= \pi \int_0^6 \frac{4}{9}(6 - y)^2 dy$$

$$= \frac{4\pi}{9} \left[\frac{(6 - y)^3}{-3} \right]_0^6$$

$$= -\frac{4\pi}{27} (0 - (6)^3)$$

$$Volume = \frac{16\pi}{3} \text{ units}^3$$

[Maximum mark: 7]

A continuous random variable X has probability density function given by

$$f(x) = \begin{cases} \frac{2(5-x)}{25}, & 0 \leq x \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the mean of X . [2](b) Find the median of X . [3]

(c) Verify that the mean is greater than the median. [2]

Answer:

$$\begin{aligned} (a) \quad E(x) &= \int_0^5 x \left[\frac{2(5-x)}{25} \right] dx \\ &= \frac{2}{25} \int_0^5 (5x - x^2) dx \\ &= \frac{2}{25} \left[\frac{5}{2}x^2 - \frac{x^3}{3} \right]_0^5 \\ &= \frac{2}{25} \left[\frac{5}{2}(5)^2 - \frac{(5)^3}{3} \right] \end{aligned}$$

$$E(x) = \frac{5}{3}$$

(b) Let m be median

$$\therefore \int_0^m f(x) dx = \frac{1}{2}$$

$$\int_0^m \frac{2(5-x)}{25} dx = \frac{1}{2}$$

$$\frac{2}{25} \left[\left(5x - \frac{x^2}{2} \right) \right]_0^m = \frac{1}{2}$$

$$5(m) - \frac{m^2}{2} = \frac{25}{4}$$

$$20m - 2m^2 = 25$$

$$2m^2 - 20m + 25 = 0$$

$$m = \frac{20 \pm \sqrt{400 - 4(2)(25)}}{2(2)}$$

$$m = 5 \pm \frac{5}{2}\sqrt{2}$$

$$m = 5 + \frac{5}{2}\sqrt{2} \quad \text{or} \quad m = 5 - \frac{5}{2}\sqrt{2}$$

Since $0 \leq m \leq 5$ We reject $m = 5 + \frac{5}{2}\sqrt{2}$

$$\therefore m = 5 - \frac{5\sqrt{2}}{2}$$

(C) $median = 5 - \frac{5\sqrt{2}}{2} \approx 1.464466 \dots$ and $mean = \frac{5}{3} \approx 1.6666 \dots$

$$\therefore Mean > median$$

[Maximum mark: 7]

Given that $\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta$, show that

$$\cos \frac{\pi}{10} = \sqrt{\frac{5 + \sqrt{5}}{8}}$$

Answer: Let $\theta = \frac{\pi}{10}$

$$\therefore \cos \frac{\pi}{2} = 16 \cos^5 \left(\frac{\pi}{10} \right) - 20 \cos^3 \left(\frac{\pi}{10} \right) + 5 \cos \left(\frac{\pi}{10} \right)$$

$$\text{Since } \cos \frac{\pi}{2} = 0$$

$$\therefore 16 \cos^5 \left(\frac{\pi}{10} \right) - 20 \cos^3 \left(\frac{\pi}{10} \right) + 5 \cos \left(\frac{\pi}{10} \right) = 0$$

$$\therefore \cos \left(\frac{\pi}{10} \right) \left[16 \cos^4 \left(\frac{\pi}{10} \right) - 20 \cos^2 \left(\frac{\pi}{10} \right) + 5 \right] = 0$$

$$\text{Since } \cos \left(\frac{\pi}{10} \right) \neq 0$$

$$\therefore 16 \cos^4 \left(\frac{\pi}{10} \right) - 20 \cos^2 \left(\frac{\pi}{10} \right) + 5 = 0$$

$$\text{Let } u = \cos^2 \left(\frac{\pi}{10} \right)$$

$$\therefore 16u^2 - 20u + 5 = 0$$

$$u = \frac{20 \pm \sqrt{20^2 - 4(16)(5)}}{2(16)}$$

$$u = \frac{20 \pm \sqrt{80}}{32} = \frac{5 \pm \sqrt{5}}{8}$$

$$\cos^2 \left(\frac{\pi}{10} \right) = \frac{5 \pm \sqrt{5}}{8}$$

$$\cos\left(\frac{\pi}{10}\right) = \pm \sqrt{\frac{5 \pm \sqrt{5}}{8}}$$

So there are 4 possible values for $\cos\left(\frac{\pi}{10}\right)$

But we need to show that one correct solution is $\sqrt{\frac{5 + \sqrt{5}}{8}}$

Now $\frac{\pi}{10}$ is in the first quadrant, because $0 < \frac{\pi}{10} < \frac{\pi}{2}$

where $\cos \theta > 0$

This eliminates two possibilities, we now have

$$\cos\left(\frac{\pi}{10}\right) = \sqrt{\frac{5 \pm \sqrt{5}}{8}}$$

now comparing $\cos\left(\frac{\pi}{10}\right)$ With $\cos\left(\frac{\pi}{6}\right)$

$$\frac{\pi}{10} < \frac{\pi}{6} \Rightarrow \cos\left(\frac{\pi}{10}\right) > \cos\left(\frac{\pi}{6}\right)$$

As Cosine function is decreasing in the first quadrant

$$\therefore \sqrt{\frac{5 \pm \sqrt{5}}{8}} > \frac{\sqrt{3}}{2}$$

$$\frac{5 \pm \sqrt{5}}{8} > \frac{3}{4}$$

$$5 \pm \sqrt{5} > 6$$

$$\text{but } 5 - \sqrt{5} < 6$$

$$\therefore \cos\left(\frac{\pi}{10}\right) \neq \sqrt{\frac{5 - \sqrt{5}}{8}}$$

Hence

$$\cos\left(\frac{\pi}{10}\right) = \sqrt{\frac{5 + \sqrt{5}}{8}}$$



[Maximum mark: 14]

The function f is defined for $x \in \mathbb{R}$ by

$$f(x) = e^{4x} + 1$$

(a) (i) Write down the range of f .(ii) Write down the equation of the horizontal asymptote of the graph of f .

[2]

(b) Find $f^{-1}(x)$.

[3]

(c) On the same set of axes, sketch the graphs of $y = f(x)$ and $y = f^{-1}(x)$. Clearly indicate the asymptotes of each graph.

[3]

The function g is defined for $x \in \mathbb{R}$ by

$$g(x) = 5x - 2$$

(d) Find $(g \circ f)(x)$, giving your answer in its simplest form.

[2]

(e) Hence solve the equation $(g \circ f)(x) = 7$. Give your answer in the form $a \ln b$, where $a, b \in \mathbb{Q}^+$.

[4]

Answer:

(a) Since

$$(i) \quad e^{4x} > 0 \Rightarrow f(x) > 1$$

The range is $f(x) > 1$ (ii) As $x \rightarrow -\infty, e^{4x} \rightarrow 0$

$$f(x) = e^{4x} + 1$$

So, the graph approaches the line $y = 1$ ✓

$$(b) \quad f(x) = e^{4x} + 1$$

$$\therefore y = e^{4x} + 1$$

$$x = e^{4y} + 1$$

$$\Rightarrow x - 1 = e^{4y}$$

$$\Rightarrow \ln(x - 1) = 4y$$

$$\Rightarrow y = \frac{1}{4} \ln |x - 1|$$

$$\therefore f^{-1}(x) = \frac{1}{4} \ln |x - 1|$$

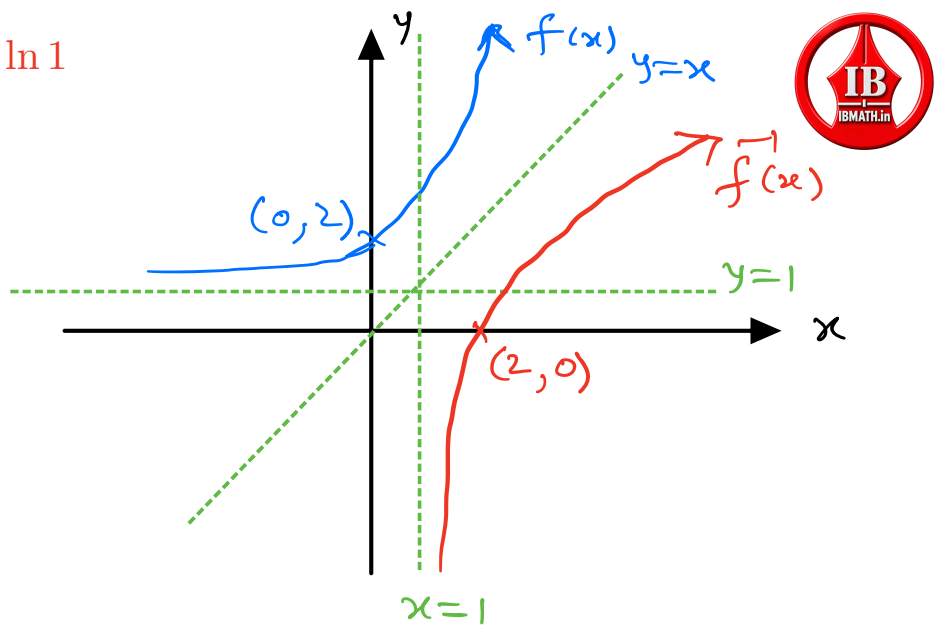
$$(c) \quad \ln|x-1| = 0 = \ln 1$$

$$x-1 = 1$$

$$x = 2$$

Asymptotes are

$$y=1, \quad x=2$$



$$(d) \quad g(x) = 5x - 2$$

$$(g \circ f)(x) = g(f(x)) = g(e^{4x} + 1) = 5(e^{4x} + 1) - 2$$

$$\therefore (g \circ f)(x) = 5e^{4x} + 3$$

$$(e) \quad (g \circ f)(x) = 7$$

$$\Rightarrow 5e^{4x} + 3 = 7$$

$$5e^{4x} = 4$$

$$e^{4x} = \frac{4}{5}$$

$$x = \frac{1}{4} \ln \left(\frac{4}{5} \right)$$

[Maximum mark: 19]

Consider the function f , defined by $f(x) = \frac{x \ln(1 - x^2)}{\sqrt{1 - x^2}}$.

(a) Determine the domain of f . [2]

(b) Show that f is an odd function. [1]

(c) By using the substitution $x = \sin \theta$, show that

$$\int_0^1 f(x) dx = 2 \int_0^{\frac{\pi}{2}} \sin \theta \ln(\cos \theta) d\theta$$

[4]

(d) Using l'Hôpital's rule, find

$$\lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\ln(\cos \theta)}{\sec \theta}$$

[4]

(e) Hence, find $\int_0^1 f(x) dx$. [6]

(f) State the value of $\int_{-1}^1 f(x) dx$. Justify your answer. [2]

Answer:

(a) Domain is

$$1 - x^2 > 0$$

$$x^2 < 1$$

$$\pm x < 1$$

$$\therefore x < 1, x > -1 \Rightarrow -1 < x < 1$$

$$(b) f(x) = \frac{x \ln(1-x^2)}{\sqrt{1-x^2}}.$$

$$f(-x) = \frac{(-x) \ln(1-(-x)^2)}{\sqrt{1-(-x)^2}}$$

$$f(-x) = -\frac{x \ln(1-x^2)}{\sqrt{1-x^2}}$$

$$\therefore f(-x) = -f(x)$$

Hence f is an odd function

(c)

$$f(x) = \frac{x \ln(1-x^2)}{\sqrt{1-x^2}}.$$

$$\int_0^1 f(x) dx =$$

put $x = \sin \theta \Rightarrow dx = \cos \theta d\theta$

and $x^2 = \sin^2 \theta$

$$1-x^2 = 1-\sin^2 \theta = \cos^2 \theta$$

$$\therefore \sqrt{1-x^2} = \cos \theta$$

When $x = 1 \Rightarrow \sin \theta = 1 \Rightarrow \theta = \frac{\pi}{2}$

and $x = 0 \Rightarrow \theta = 0$

$$\int_0^1 f(x) dx = \int_0^{\pi/2} \frac{(\sin \theta) \ln(\cos^2 \theta)}{\cos \theta} (\cos \theta d\theta)$$

$$\therefore \int_0^1 f(x) dx = 2 \int_0^{\pi/2} \sin \theta \ln(\cos \theta) d\theta \quad (\because \ln a^b = b \ln a)$$

(d) $L = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\ln(\cos \theta)}{\sec \theta}$ Using l'Hôpital's rule

Since this limit is not defined at $\theta = \frac{\pi}{2}$

$$L = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\frac{1}{\cancel{\cos \theta}} \times (-\sin \theta)}{\cancel{\sec \theta} \tan \theta}$$

$$= \lim_{\theta \rightarrow \frac{\pi}{2}} (-\cos \theta) = 0$$

$$\therefore \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\ln(\cos \theta)}{\sec \theta} = 0$$

(e) $\int_0^1 f(x) dx = 2 \int_0^{\frac{\pi}{2}} \sin \theta \ln(\cos \theta) d\theta$

$$= 2 \int_0^{\pi/2} \ln(\cos \theta) \cdot \sin \theta d\theta$$

(u) ($\frac{dv}{dx}$)

(ILATE)

using formula 5.16

AHL 5.16	Integration by parts	$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$ or $\int u dv = uv - \int v du$
---------------------------	----------------------	--

$$\int_0^1 f(x) dx = 2 [\ln(\cos \theta) (-\cos \theta)]_0^{\pi/2} - 2 \int_0^{\pi/2} \frac{1}{\cancel{\cos \theta}} \cdot (-\sin \theta) \cdot (-\cancel{\cos \theta}) d\theta$$

as $x \rightarrow \frac{\pi}{2}$

$$= 2 [-\cos \theta \ln(\cos \theta)] \rightarrow 2 [-(0) + (0)] = 0$$

$$\therefore \int_0^1 f(x) dx = 0 - 2 \int_0^{\pi/2} \sin \theta d\theta$$

$$= 2 [\cos \theta]_0^{\pi/2} = 2 \left(\cos \frac{\pi}{2} - \cos 0 \right) = 2 (0 - 1) = -2$$

$$\therefore \int_0^1 f(x) dx = -2$$

(f) Since $f(x)$ is an odd function

$$\therefore \int_{-1}^1 f(x) dx = \int_{-1}^0 f(x) dx + \int_0^1 f(x) dx$$

In first term, let $x = -x$, $f(-x) \rightarrow -f(x)$

$$\therefore \int_{-1}^0 f(x) dx = - \int_0^1 f(x) dx$$

$$\therefore \int_{-1}^1 f(x) dx = - \int_0^1 f(x) + \int_0^1 f(x) = 0$$

Question 12

[Maximum mark: 19]

The points $A(1, 0, 0)$, $B(0, 1, 0)$ and $C(0, 0, 1)$ lie in the plane Π_1 .

(a) Find the Cartesian equation of the plane Π_1 . [4]

(b) Show that the shortest distance from the origin O to the plane Π_1 is $\frac{1}{\sqrt{3}}$. [3]

A second plane is defined by

$$\Pi_2 : x - y + z = -1$$

(c) Find a vector equation of the line L where Π_1 and Π_2 intersect. [4]

A third plane is defined by

$$\Pi_3 : 2x + y - \alpha z = \beta$$

It is given that planes Π_1 , Π_2 and Π_3 intersect at a unique point.

(d) Find the coordinates of the point of intersection of Π_1 , Π_2 and Π_3 in terms of α and β . [4]

The distance from the origin O to this point of intersection is three times the shortest distance from O to Π_1 .

(e) Find the possible values of β in terms of α . [4]

Answer:

$$(a) \quad \vec{AB} = \begin{pmatrix} 0-1 \\ 1-0 \\ 0-0 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$$

$$\vec{AC} = \begin{pmatrix} 0-1 \\ 0-0 \\ 1-0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$\therefore$ Normal to plane Π_1 is $\vec{AB} \times \vec{AC}$

$$\vec{n}_1 = \begin{vmatrix} i & j & k \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = i + j + k$$

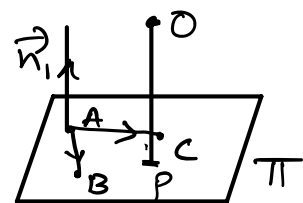
$$\vec{a}_0 = \vec{OA} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$\therefore$ eqn. of plane Π_1 is

$$\vec{n}_1 \cdot \vec{r} = \vec{n}_1 \cdot \vec{a}_0$$

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\boxed{x + y + z = 1} \rightarrow \Pi_1$$



(b)

(b) Show that the shortest distance from the origin O to the plane Π_1 is $\frac{1}{\sqrt{3}}$.Shortest distance from O to π_1 Let it be OP $\therefore \overrightarrow{OP}$ is in direction of $\vec{n}_1$ $\therefore$ equation of $\vec{OP}$ is

$$\vec{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

$$(\vec{r} = \vec{a} + \lambda \vec{b})$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \lambda \\ \lambda \\ \lambda \end{pmatrix}$$

 $\therefore (\lambda, \lambda, \lambda)$ are coords. of P which lies on π_1

$$x + y + z = 1$$

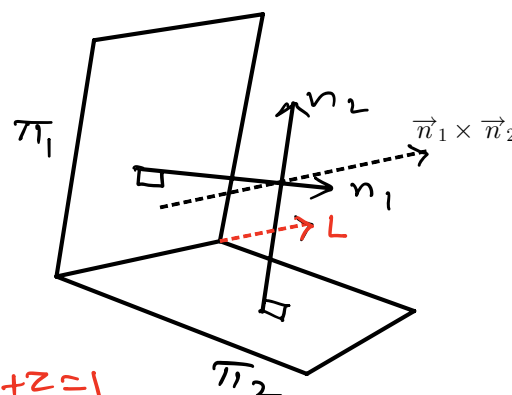
$$\therefore \begin{aligned} \lambda + \lambda + \lambda &= 1 \\ 3\lambda &= 1 \Rightarrow \lambda = \frac{1}{3} \end{aligned}$$

$$\therefore OP = |\overrightarrow{OP}| = \left| \begin{pmatrix} 1/3 \\ 1/3 \\ 1/3 \end{pmatrix} \right| = \sqrt{\frac{1}{9} + \frac{1}{9} + \frac{1}{9}} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

therefore, Shortest distance from Origin to plane π_1 , is $\frac{1}{\sqrt{3}}$ (c) Find a vector equation of the line L where Π_1 and Π_2 intersect.Any vector parallel to π_1 is perpendicular to its normal vector $\vec{n}_1$ Similarly, any vector parallel to π_2 is perpendicular to its normal vector $\vec{n}_2$ A line of intersection of π_1 & π_2 has direction vector that is parallel to both planes, and so is perpendicular to $\vec{n}_1$ & $\vec{n}_2$

$$\vec{n}_1 = i + j + k$$

$$\text{and } \vec{n}_2 = i - j + k$$



$$\pi_1 : x + y + z = 1$$

$$\pi_2 : x - y + z = -1$$

Now common perpendicular to n_1 & n_2 is

$$\vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} i & j & k \\ 1 & 1 & 1 \\ 1 & -1 & 1 \end{vmatrix} = 2i - 2k = 2 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

Since equation of plane π_1 & π_2 are

$$\pi_1 : x + y + z = 1$$

$$\pi_2 : x - y + z = -1$$

Let $z = 0 \quad \therefore x + y = 1$

& $x - y = -1$

$$2x = 0 \Rightarrow x = 0 \quad \therefore y = 1$$

$\therefore (0, 1, 0)$ Is a point on line L

Therefore, equation of line L is

$$\vec{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$(\vec{r} = \vec{a} + t\vec{b})$$

It is given that planes Π_1 , Π_2 and Π_3 intersect at a unique point.

(d) Find the coordinates of the point of intersection of Π_1 , Π_2 and Π_3 in terms of α and β . [4]

$$\pi_1 : x + y + z = 1$$

$$\pi_2 : x - y + z = -1$$

$$\Pi_3 : 2x + y - \alpha z = \beta$$

From part(c), we know that unique point of intersection must lie on line L

Therefore, coordinates of common point intersection $(t, 1, -t)$

Substituting these coordinates of common point in π_3 , we get

$$2t + 1 + \alpha t = \beta$$

$$\therefore 2t + \alpha t = \beta - 1$$

$$\left\{ \vec{r} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} t \\ 1 \\ -t \end{pmatrix} \right\}$$

(from part (c))

$$\therefore t(2 + \alpha) = \beta - 1$$

$$t = \frac{\beta - 1}{2 + \alpha}$$

coordinates of common point of intersection are

$$\left(\frac{\beta - 1}{2 + \alpha}, 1, -\frac{\beta - 1}{2 + \alpha} \right), \alpha \neq -2$$

The distance from the origin O to this point of intersection is three times the shortest distance from O to Π_1 .

(e) Find the possible values of β in terms of α .

[4]

(e) Let d , be the distance from origin to this point

$$\therefore d = 3(OP) = 3 \left(\frac{1}{\sqrt{3}} \right) = \sqrt{3}$$

$$\therefore \sqrt{\left(\frac{\beta - 1}{2 + \alpha} - 0 \right)^2 + (1 - 0)^2 + \left(-\frac{\beta - 1}{2 + \alpha} - 0 \right)^2} = \sqrt{3}$$

$$\left(\frac{\beta - 1}{2 + \alpha} \right)^2 + 1 + \left(\frac{\beta - 1}{2 + \alpha} \right)^2 = 3$$

$$2 \left(\frac{\beta - 1}{2 + \alpha} \right)^2 = 2$$

$$\therefore \frac{\beta - 1}{2 + \alpha} = \pm 1$$

$$\beta - 1 = (2 + \alpha) \quad \text{and} \quad \beta - 1 = -(2 + \alpha)$$

$$\therefore \beta = 3 + \alpha, \alpha \neq -2$$

$$\text{and } \beta = -\alpha - 1, \alpha \neq -2$$

Request from teacher:

If you really like my method of teaching, then please like, share and subscribe to my YouTube channel.

*Also, click bell icon to get my new **LIVE** videos.*

End of Lesson

Contact me on:  **9326363793**

You can download the PDF of this lesson from the link, given in the description of this video. My channel link is clickable at the bottom of the all pages.



$\int f(x) dx$

$\int f(x) dx$

$\sin x$

$a^2 + b^2 = c^2$

$f(x)$

$f(x)$

IB Mathematics

AAHL PAPER-2

PREDICTION EXAM

for November 2026

with SOLUTION

Prepare Practice Perform!

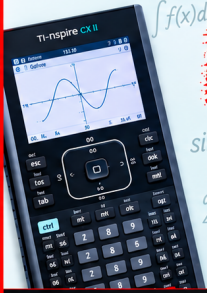
$a^2 + b^2 = c^2$

$f(x)$

☒ Past Paper Trends

☒ Important Topics

☒ Exam Style Questions





IB MATHEMATICS

Analysis and Approaches

Applications and Interpretation

Higher Level / Standard Level

IB Mathematics Examination

Your Success Our Goal!

 [youtube.com/@IBMATH-in](https://www.youtube.com/@IBMATH-in)

 **9326363793**